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GRADO EN FÍSICA

Práctica de Complejos

Quantum Chaos and the Statistics of Energy Levels

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Abstract

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1 Introduction

The study of complex systems allows to explore the transition between the classical order and the statistical manifestations in the quantum regime. As in classical mechanics chaos is defined as the exponential sensitivity to initial conditions, the linearity in the Schrödinger equation does not allow for a direct translation of this definition to quantum mechanics. However, the phenomena known as 'Quantum Chaos' is manifested throughout the statistical properties of the energy spectrum of the system.

The nuclei in this project is the Bohigas-Giannoni-Schmit (BGS) conjecture, which states that the energy levels of a quantum system whose classical counterpart is chaotic will exhibit the same statistical properties as the eigenvalues of a random matrix ensemble.

To validate this hypothesis, we'll investigate the distribution of spacing between adjacent energy levels in two systems (s_i): one integrable and one chaotic.

The first one is the two-dimensional **rectangular billiard**, which is an **integrable system**. In this system, the random matrix theory (RMT) predicts that the distribution of level spacings follows a Poisson distribution, which is given by:

$$P(s) = e^{-s} \quad (1.1)$$

The second system is the **stadium billiard**, which is a **chaotic system**. In this case, RMT predicts that the distribution of level spacings follows the Wigner-Dyson distribution, which is given by:

$$P(s) = \frac{\pi}{2} s e^{-\frac{\pi}{4} s^2} \quad (1.2)$$

where s is the normalized spacing between adjacent energy levels in both cases.

To integrate the time independent Schrödinger equation for these systems, we will use the **Finite Difference Method** (FDM): this method allows us to discretize the spatial domain and solve for the energy eigenvalues and eigenfunctions of the system.

2 Methods

To be able to solve the Schrödinger's Equation $H\psi = E\psi$, we'll first need to discretize the space and then solve the resulting eigenvalue problem. We'll need to impose the Dirichlet conditions to the domain, which means that the wavefunction ψ must vanish at the boundaries of the billiard.

$$\psi|_{\partial\Omega} = 0 \quad (2.1)$$

All the methods used in this project are implemented in Python, and the code is available in the GitHub repository . To explain the followed procedure, we'll first introduce the Mascara method, which is used to discretize the system, and then the Finite Difference Method, which is used to solve the resulting eigenvalue problem.

2.1 Discretisation of the system: Mascara Method

To easily create different geometries, we decided to implement a finite difference method that takes different masks and computes the corresponding eigenvalues. The mask represents a grid containing the geometry of the problem. For each pixel, we determine whether it belongs to the geometry by assigning it a true or false value.

To carry out this project, we created three different masks: a rectangular mask, a stadium-shaped mask, and a quarter-stadium mask. The following figure shows these masks:

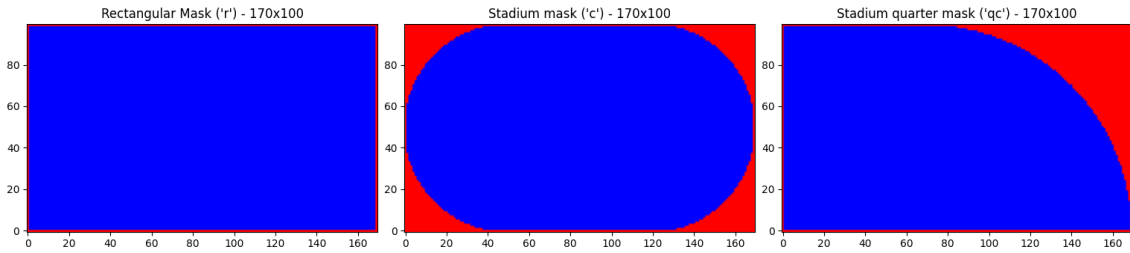


Figura 1: Different mask geometries used in this experiment. From left to right: a rectangular mask, a stadium-shaped mask, and a quarter-stadium mask.

The project aimed to investigate the stadium geometry, which produces chaotic behavior in the eigenvalue spectrum. However, we encountered a problem: the stadium has two symmetries, one with respect to the x -axis and another with respect to the y -axis. In Chapter 1.4 of the thesis by *Wisniacki, Diego A.*, it is explained that the eigenvalues are divided into four different sets because of these symmetries:

- even | even
- even | odd
- odd | even
- odd | odd

This produces four different Wigner–Dyson distributions, which, when combined, result in a Poisson distribution. By removing the symmetries and considering only a quarter of the stadium, the chaotic behavior becomes visible. [1]

2.2 Finite Difference Method

Once we have a grid of the system, where $x_i = i\Delta x$ and $y_j = j\Delta y$, being $i = 1, 2, \dots, n_x$ and $j = 1, 2, \dots, n_y$. As we know, the HAmiltonian for a two-dimensional system is given by:

$$H = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + V(x, y) \quad (2.2)$$

Since in our systems we have a free particle, the potential $V(x, y)$ is zero everywhere inside the billiard and infinite outside.

3 Results

4 Conclusions

Referencias

- [1] Diego Ariel Wisniacki. *Caos cuántico en sistemas dependientes de un parámetro*. PhD thesis, Universidad de Buenos Aires. Facultad de Ciencias Exactas y Naturales, 1999.